

Homework Assignment 1: Slab-Geometry Boltzmann Equation

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Question 1 - Useful identity for the solution of the time-dependent slab-geometry Boltzmann equation

In this question we are asked to prove the following identity:

$$\psi(x, \mu, t; c) = ce^{-(1-c)v\Sigma_t t} \psi(cx, \mu, ct; 1) \quad (1)$$

We will do it in two parts. First, we will write down the equation that $\tilde{\psi} = \psi(cx, \mu, ct; 1)$ satisfies, and then we will use it in order to prove that the product $ce^{-(1-c)v\Sigma_t t} \tilde{\psi}$ satisfies the same equation as $\psi(x, \mu, t; c)$.

The equation for $\tilde{\psi}$

Notation: We will denote by $\Psi(X, \mu, T) \equiv \psi(X, \mu, T; 1)$ the $c = 1$ solution written in its own variables, and by Ψ_T and Ψ_X its partial derivatives with respect to them, so that $\tilde{\psi}(x, \mu, t) = \Psi(cx, \mu, ct)$ with $X = cx$ and $T = ct$. **This notation is used to avoid confusion when using the chain rule, as will be done immediately.** First, Ψ satisfies the $c = 1$ equation in the variables X and T :

$$\frac{1}{v} \Psi_T + \mu \Psi_X + \Sigma_t \Psi = \frac{\Sigma_t}{2} \int_{-1}^1 \Psi d\mu' \quad (2)$$

which after applying the chain rule, $\Psi_T = \frac{1}{c} \frac{\partial \tilde{\psi}}{\partial t}$ and $\Psi_X = \frac{1}{c} \frac{\partial \tilde{\psi}}{\partial x}$, and multiplying by c , becomes:

$$\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c \Sigma_t \tilde{\psi} = \frac{c \Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' \quad (3)$$

So $\tilde{\psi}$ obeys the same equation as $\psi(x, \mu, t; c)$ except that it is removed at the rate $c \Sigma_t$ instead of Σ_t .

Proving the identity

In this part we are going to simply substitute the RHS of 1 into the LHS of the equation for $\psi(x, \mu, t; c)$ and show that it satisfies the same equation. After substitution and application of the chain rule we

get:

$$\text{LHS} = \frac{1}{v} \left[c(c-1)v\Sigma_t e^{-(1-c)v\Sigma_t t} \tilde{\psi} + c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial t} \right] + \mu c e^{-(1-c)v\Sigma_t t} \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t c e^{-(1-c)v\Sigma_t t} \tilde{\psi} \quad (4)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + \Sigma_t \tilde{\psi} + (c-1)\Sigma_t \tilde{\psi} \right] \quad (5)$$

$$= c e^{-(1-c)v\Sigma_t t} \left[\frac{1}{v} \frac{\partial \tilde{\psi}}{\partial t} + \mu \frac{\partial \tilde{\psi}}{\partial x} + c\Sigma_t \tilde{\psi} \right] \quad (6)$$

$$= c e^{-(1-c)v\Sigma_t t} \frac{c\Sigma_t}{2} \int_{-1}^1 \tilde{\psi} d\mu' = \frac{c\Sigma_t}{2} \int_{-1}^1 c e^{-(1-c)v\Sigma_t t} \tilde{\psi} d\mu' \quad (7)$$

where 3 was used in the last line, and this is exactly the RHS of the equation for $\psi(x, \mu, t; c)$.

Question 2 - Derivation of the time-dependent planar scalar flux for a plane source

The relation between a plane source and a point source as studied in class is given by:

$$\phi_{\text{pl}}(x) = 2\pi \int_{|x|}^{\infty} \phi_{\text{pt}}(r) r dr \quad (8)$$

The solution taken from J. C. J. Paasschens is the point-source scalar flux $\phi_{\text{pt}}(r, t)$ for $c = 1$, and therefore the planar-source scalar flux is given by the integration given above.

$$\phi_{\text{pl}}(x, t) = 2\pi \int_{|x|}^{\infty} \frac{e^{-v\Sigma_t t}}{4\pi r^2} \delta(r - vt) r dr + 2\pi \int_{|x|}^{\infty} \frac{(1 - r^2/v^2 t^2)^{1/8}}{(4\pi vt/(3\Sigma_t))^{3/2}} e^{-v\Sigma_t t} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) \Theta(vt - r) r dr \quad (9)$$

$$= \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-v\Sigma_t t}}{(4\pi vt/(3\Sigma_t))^{3/2}} \int_{|x|}^{vt} \left(1 - \frac{r^2}{v^2 t^2}\right)^{1/8} G\left(v\Sigma_t t \left[1 - \frac{r^2}{v^2 t^2}\right]^{3/4}\right) r dr \quad (10)$$

$$\stackrel{\xi=r/vt}{=} \frac{e^{-v\Sigma_t t}}{2vt} \Theta(vt - |x|) + \Theta(vt - |x|) \frac{2\pi e^{-v\Sigma_t t} (vt)^{1/2}}{(4\pi/(3\Sigma_t))^{3/2}} \int_{\frac{|x|}{vt}}^1 (1 - \xi^2)^{1/8} G\left(a [1 - \xi^2]^{3/4}\right) \xi d\xi \quad (11)$$

$$\stackrel{u=1-\xi^2}{\stackrel{w=au^{3/4}}{=}} \frac{e^{-v\Sigma_t t}}{2vt} \left[1 + \sqrt{\frac{3}{\pi}} \int_0^{w_0} \sqrt{w} G(w) dw \right] \Theta(vt - |x|) \quad (12)$$

with $w_0 = a \left(1 - \frac{x^2}{(vt)^2}\right)^{3/4}$, the powers combining as $\frac{1}{6} + \frac{1}{3} = \frac{1}{2}$ and the Σ_t dependence cancelling in the last step (I used the assistance of AI in this algebraic derivation). Finally, with the interpolation formula of Paasschens, $G(w) = e^w \sqrt{1 + b/w}$ with $b = 2.026$, the integral has a closed form:

$$\int_0^{w_0} e^w \sqrt{w + b} dw = e^{w_0} \sqrt{w_0 + b} - \sqrt{b} - \frac{\sqrt{\pi}}{2} e^{-b} \left[\text{erfi}\left(\sqrt{w_0 + b}\right) - \text{erfi}\left(\sqrt{b}\right) \right] \quad (13)$$

which **closes the derivation of the planar-source scalar flux for $c = 1$** . Of course the relation proved in Q1 can be used to get the solution for $c \neq 1$ using a simple scaling and multiplication by an exponential factor of the solution for $c = 1$.

Question 3 - Classical and asymptotic diffusion against the exact solution

Derivation of the time-dependent diffusion Green's function

The time-dependent diffusion equation is:

$$\frac{1}{v} \frac{\partial \phi}{\partial t} - D \frac{\partial^2 \phi}{\partial x^2} + \Sigma_a \phi = S(x, t) \quad (14)$$

Take $S = \delta(x)\delta(t)$ and multiply (14) by v :

$$\frac{\partial \phi}{\partial t} - Dv \frac{\partial^2 \phi}{\partial x^2} + v\Sigma_a \phi = v\delta(x)\delta(t) \quad (15)$$

Step 1: the pulse source is an initial condition. Integrate (15) over $t \in (0^-, 0^+)$. The diffusion and absorption terms are bounded and contribute nothing in the limit, so only the time derivative survives:

$$\phi(x, 0^+) - \phi(x, 0^-) = v\delta(x) \quad \implies \quad \phi(x, 0^+) = v\delta(x) \quad (16)$$

since nothing precedes the pulse. For $t > 0$ the equation is source-free, so the problem is now a pure initial-value problem (this is the same trick as in Assignment 1, Question 2)

Step 2: absorption factors out exactly. As performed in the course, substituting $\phi(x, t) = e^{-v\Sigma_a t} u(x, t)$ into (15) for $t > 0$, the $-v\Sigma_a u$ produced by the time derivative cancels the absorption term identically and leaves the pure heat equation

$$\frac{\partial u}{\partial t} = Dv \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0^+) = v\delta(x) \quad (17)$$

Note this step is untouched by the sign of $1 - c$: for $c > 1$ the exponential grows instead of decays, and nothing else changes.

Step 3: Fourier transform. With $\hat{u}(k, t) = \int_{-\infty}^{\infty} u(x, t) e^{-ikx} dx$, the spatial derivative becomes multiplication by $-k^2$ and the PDE becomes an ordinary differential equation in t for each k separately:

$$\frac{d\hat{u}}{dt} = -Dvk^2 \hat{u}, \quad \hat{u}(k, 0^+) = v \quad \implies \quad \hat{u}(k, t) = v e^{-Dvk^2 t} \quad (18)$$

the initial value being v because the transform of $v\delta(x)$ is the constant v - a delta excites every wavenumber equally.

Step 4: invert. Completing the square in the exponent, $-Dvt k^2 + ikx = -Dvt \left(k - \frac{ix}{2Dvt}\right)^2 - \frac{x^2}{4Dvt}$, and using $\int_{-\infty}^{\infty} e^{-\alpha k^2} dk = \sqrt{\pi/\alpha}$:

$$u(x, t) = \frac{v}{2\pi} \int_{-\infty}^{\infty} e^{-Dvk^2 t} e^{ikx} dk = \frac{v}{\sqrt{4\pi Dvt}} e^{-x^2/(4Dvt)} \quad (19)$$

Restoring the factor from Step 2 gives the Green's function:

$$\boxed{\phi_{\text{diff}}(x, t; c) = \frac{v e^{-(1-c)\Sigma_a t}}{\sqrt{4\pi Dvt}} \exp\left(-\frac{x^2}{4Dvt}\right)} \quad (20)$$

which in the units of this assignment, $\Sigma_t = v = 1$ and $D = D_0(c)$, is the form implemented in `src/homework2/diffusion.py`:

$$\phi_{\text{diff}}(x, t; c) = \frac{e^{-(1-c)t}}{\sqrt{4\pi D_0(c)t}} e^{-x^2/(4D_0(c)t)} \quad (21)$$

A second route: Fourier in space, then Laplace in time

Define

$$\hat{\phi}(k, t) = \int_{-\infty}^{\infty} \phi(x, t) e^{-ikx} dx, \quad \tilde{\phi}(k, s) = \int_{0-}^{\infty} \hat{\phi}(k, t) e^{-st} dt \quad (22)$$

and apply both to (15) term by term, with $\phi(x, 0^-) = 0$:

$$\frac{\partial \phi}{\partial t} \longrightarrow s \tilde{\phi} - \hat{\phi}(k, 0^-) = s \tilde{\phi} \quad (23)$$

$$-Dv \frac{\partial^2 \phi}{\partial x^2} \longrightarrow Dvk^2 \tilde{\phi} \quad (24)$$

$$v \Sigma_a \phi \longrightarrow v \Sigma_a \tilde{\phi} \quad (25)$$

$$v \delta(x) \delta(t) \longrightarrow v \quad (26)$$

so that

$$(s + Dvk^2 + v \Sigma_a) \tilde{\phi}(k, s) = v \quad (27)$$

$$\tilde{\phi}(k, s) = \frac{v}{s + Dvk^2 + v \Sigma_a} \quad (28)$$

a single simple pole at $s = -(Dvk^2 + v \Sigma_a)$, so with $\mathcal{L}^{-1}\{(s + \alpha)^{-1}\} = e^{-\alpha t}$,

$$\hat{\phi}(k, t) = v e^{-(Dvk^2 + v \Sigma_a)t} \quad (29)$$

and the Fourier inversion completes the square exactly as in Step 4:

$$\phi(x, t) = \frac{v}{2\pi} \int_{-\infty}^{\infty} e^{-(Dvk^2 + v \Sigma_a)t} e^{ikx} dk \quad (30)$$

$$= \frac{v e^{-v \Sigma_a t}}{2\pi} e^{-x^2/(4Dvt)} \int_{-\infty}^{\infty} e^{-Dvt(k - \frac{ix}{2Dvt})^2} dk \quad (31)$$

$$= \frac{v e^{-v \Sigma_a t}}{2\pi} e^{-x^2/(4Dvt)} \sqrt{\frac{\pi}{Dvt}} = \frac{v e^{-v \Sigma_a t}}{\sqrt{4\pi Dvt}} e^{-x^2/(4Dvt)} \quad (32)$$

which is (20), with $\Sigma_a = (1 - c)\Sigma_t$.

Parts 3(a) and 3(b): the comparison

Each figure fixes c and shows $t = 1, 2, 3, 4, 7, 15$. The vertical line marks the causal front $|x| = vt$, beyond which the exact solution is identically zero. Only $x \geq 0$ is plotted. All three solutions are even in x .

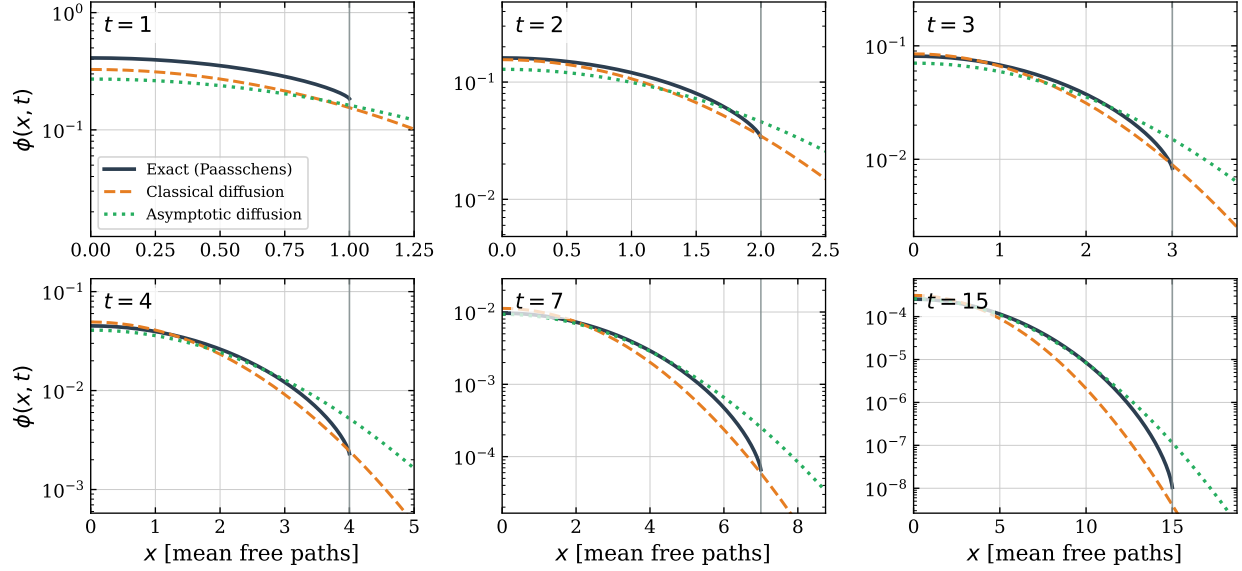


Figure 1: $c = 0.6$.

Part 3(c): the numerical solution

Before discretising, the absorption is removed exactly as in Step 2: with $\phi(x, t) = e^{-v\Sigma_a t}u(x, t)$ the term $v\Sigma_a\phi$ cancels identically and what is left is the pure heat equation carrying the pulse as its initial condition:

$$\frac{\partial u}{\partial t} = Dv \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0^+) = v \delta(x) \quad (33)$$

Since the equation and initial condition are even in x , so is the solution of (33).

Truncation. The solution is plotted in $[0, L]$ and is assumed to be zero at $x = L$. This truncation is justified because the spatial Gaussian from the analytic solution has variance $\sigma^2 = 2Dvt$, so taking

$$L = x_{\max} + 8\sqrt{2Dvt_{\max}} \quad (34)$$

places the truncation eight standard deviations beyond the last point to be plotted, where the solution is of order e^{-32} of its peak.

The grid, and the discrete pulse. Take nodes $x_j = jh$ with $j = 0 \dots N$. The spatial width is $h = L/N$, and time levels $t^n = n\Delta t$. On a node-centred grid the trapezoid weight of node 0 on the half line is $h/2$, and the full-line integral is twice the half-line one, so the cell straddling the origin carries total weight h :

$$\int_{-\infty}^{\infty} u(x, 0) dx \approx 2 \cdot \frac{h}{2} u_0^0 = h u_0^0 \stackrel{!}{=} v \quad \implies \quad u_0^0 = \frac{v}{h}, \quad u_{j \geq 1}^0 = 0 \quad (35)$$

so the spike of height v/h in the first cell is the correctly normalised delta - the factor of two from the symmetry and the half-weight of the boundary node cancel.

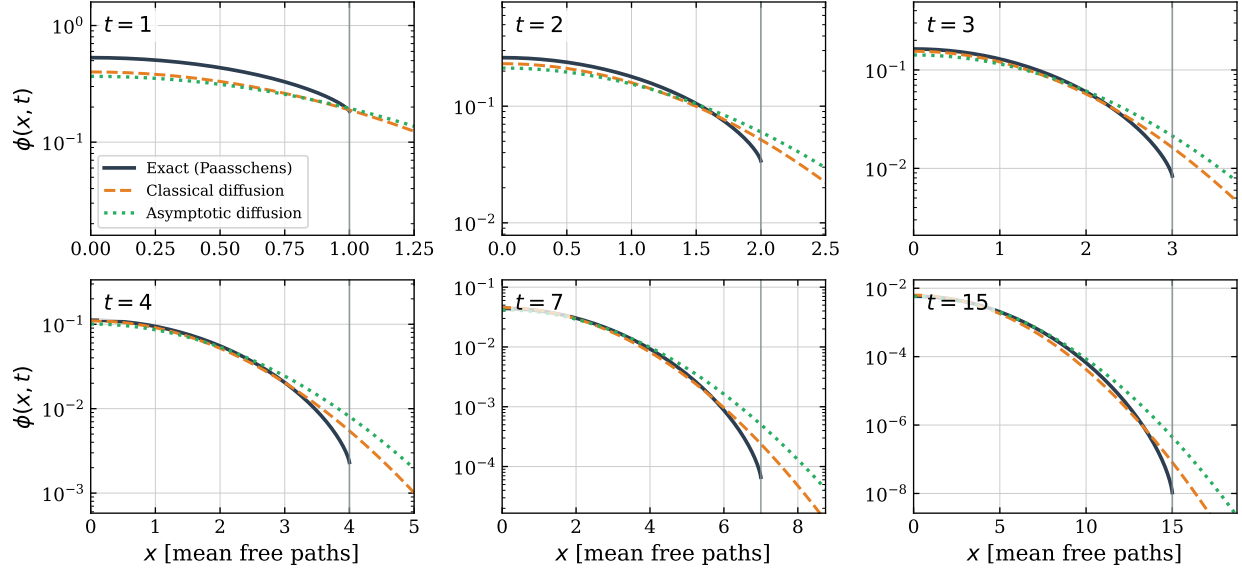


Figure 2: $c = 0.8$.

The update. Central second differences in space, evaluated at the old level, give the explicit scheme. With $r = Dv \Delta t / h^2$,

$$u_j^{n+1} = u_j^n + r(u_{j-1}^n - 2u_j^n + u_{j+1}^n), \quad 1 \leq j \leq N-1 \quad (36)$$

$$u_0^{n+1} = u_0^n + 2r(u_1^n - u_0^n) \quad (37)$$

$$u_N^{n+1} = 0 \quad (38)$$

the row at $j = 0$ following from the symmetry condition written as the ghost value $u_{-1} = u_1$.

Von Neuman Stability. Substituting the Fourier mode $u_j^n = \xi^n e^{ij\theta}$ into (36) gives the amplification factor

$$\xi(\theta) = 1 + r(e^{-i\theta} - 2 + e^{i\theta}) = 1 - 4r \sin^2(\theta/2) \quad (39)$$

and $|\xi| \leq 1$ for every θ requires $4r \leq 2$, i.e.

$$r \leq \frac{1}{2} \quad \Longleftrightarrow \quad \Delta t \leq \frac{h^2}{2Dv} \quad (40)$$

Conservation, as the check. Let M^n be the trapezoid mass on the half line, so that $\int_{-\infty}^{\infty} u dx \approx 2M^n$:

$$M^n = h \left[\frac{1}{2} u_0^n + \sum_{j=1}^{N-1} u_j^n + \frac{1}{2} u_N^n \right] \quad (41)$$

Calculating M^{n+1} using (36) and the boundary conditions gives

$$M^{n+1} - M^n = rh(u_N^n - u_{N-1}^n) = -rh u_{N-1}^n \quad (42)$$

So the scheme conserves particles to machine precision until the tail reaches the truncation, and the only leak is through the far boundary, at a rate that is directly observable as u_{N-1}^n . Restoring the prefactor, the check on the solver is $\int \phi dx = e^{-v\Sigma_a t}$, the same identity already used on the exact solution.

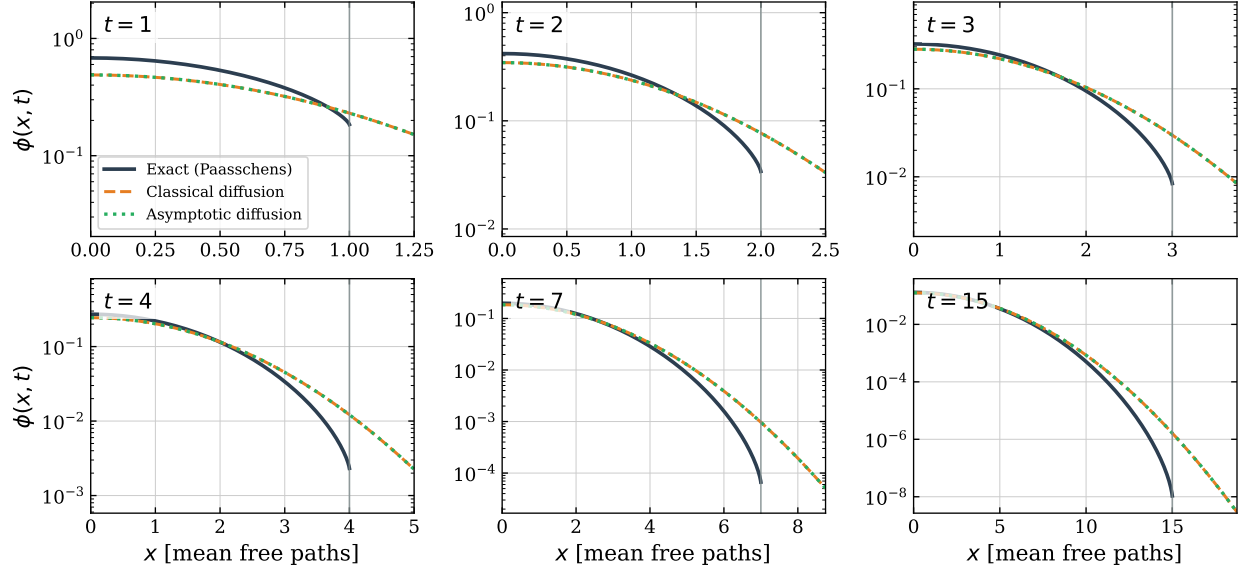


Figure 3: $c = 1$. The two diffusion coefficients coincide at $D_0 = 1/3$, so the classical and asymptotic curves overlies each other.

Results. The scheme is implemented in `src/homework2/solver.py` and run at $N = 2000$ nodes, which puts $h \approx 0.022$ and $\Delta t \approx 6.5 \times 10^{-4}$, i.e. about 2×10^4 sweeps to reach $t = 15$. Figure 6 places it over the closed form it discretises and Figure 7 measures what separates them. Three results are worth quoting:

- **Conservation is exact.** $\int \phi dx$ matches $e^{-(1-c)t}$ to between 0 and 1.1×10^{-15} over $c = 0.6, 1.0, 1.5$ and $t = 1, 4, 15$ - round-off, not discretisation, as (42) requires.
- **The scheme is second order.** The peak error at $t = 4$ falls $1.007 \times 10^{-3} \rightarrow 2.586 \times 10^{-4} \rightarrow 6.466 \times 10^{-5}$ as N doubles from 500 to 2000: ratios of 3.89 and 4.00.

Part 3(d): the numerical solutions against the exact one

Part 3(d) asks how far the numerical solution of 3(c) sits from the exact flux of Question 2. The measured quantity is the pointwise relative error

$$E(x, t; c) = \left| \frac{\phi_{\text{diff}}(x, t)}{\phi_{\text{exact}}(x, t)} - 1 \right| \quad (43)$$

where ϕ_{diff} is the march of (36) run once with $D = 1/3$ and once with the asymptotic $D = D_0(c) = (1 - c)\nu_0^2$, and ϕ_{exact} is the Question 2 flux carried to $c \neq 1$ by the scaling (1), evaluated from the exact series G .

Two choices fix the domain of Figure 8. The exact solution is identically zero beyond the causal front, so (43) is cut at $0.98vt$, just inside it. And since the front moves, the abscissa is the scaled position x/vt , which puts every time on one axis with the front always at 1.

What the figure measures. Part 3(c) established that the scheme reproduces the Green's function it discretises to 6.5×10^{-5} at $N = 2000$, while Figure 8 spans 10^{-2} to 10^4 . The discretisation

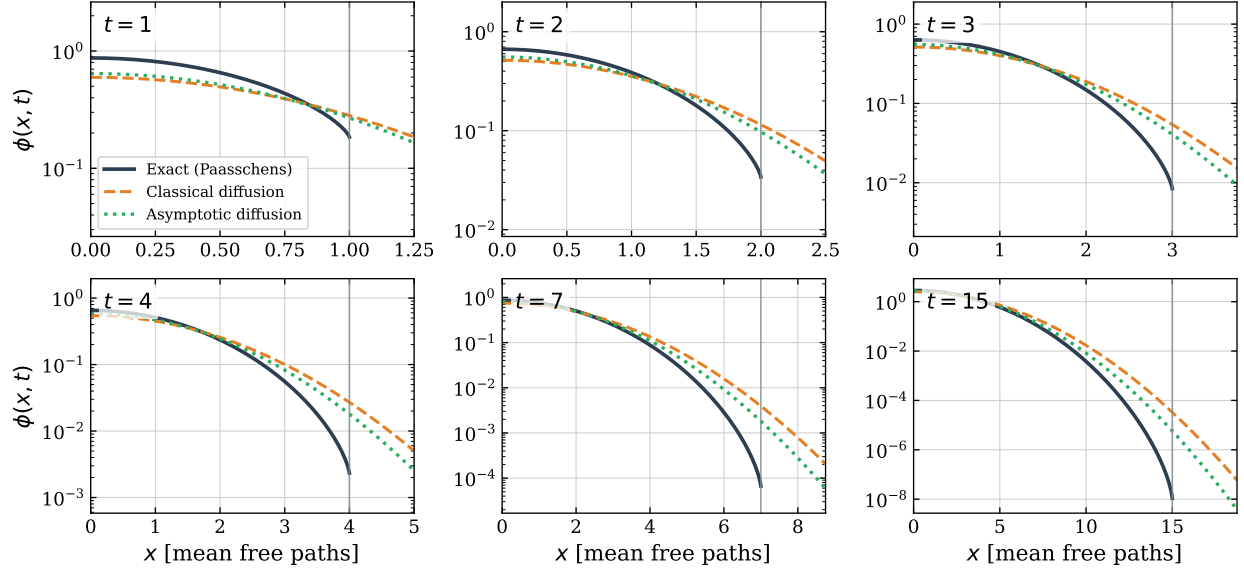


Figure 4: $c = 1.2$.

sits two to eight orders of magnitude below everything plotted, so **what is measured here is the diffusion approximation itself, not the solver**. The downward spikes are sign changes, where the error passes through zero.

Discussing the results

1. **Diffusion is a late-time limit** The relative error of both diffusion solutions decays with time, and the decay is faster the closer c is to 1. In addition, the quality of the approximation generally improves with time: the exact solution relaxes onto its diffusive shape only once a particle has scattered many times, and at $t = 1$ - one mean free time, one collision - there is nothing diffusive about it.
2. **Once relaxed, $D_0(c)$ is the better coefficient, and by a large factor.** At $t = 15$ the asymptotic error at the origin is 2.61% against 23.88% at $c = 0.6$, 0.59% against 8.39% at $c = 0.8$, 3.67% against 10.59% at $c = 1.2$ and 4.85% against 19.70% at $c = 1.5$ - a factor of four to fourteen. That is the payoff of the asymptotic coefficient: $D_0 = (1 - c)\nu_0^2$ is built from the discrete eigenvalue that governs the exact solution at long times, whereas $1/3$ is not built from anything about the medium at all.

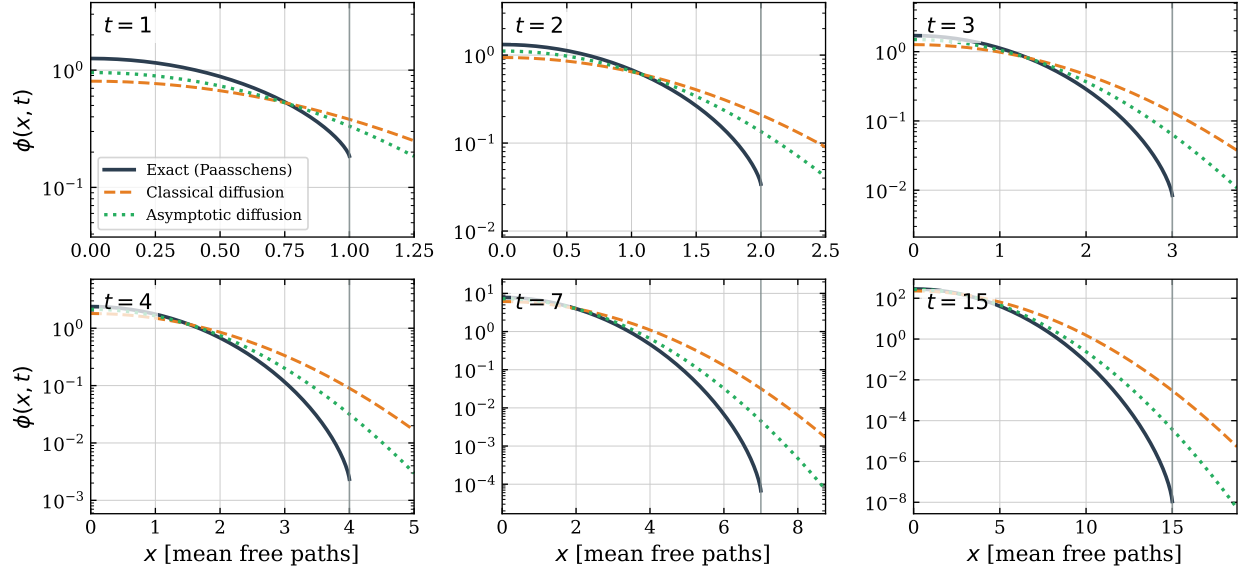


Figure 5: $c = 1.5$.

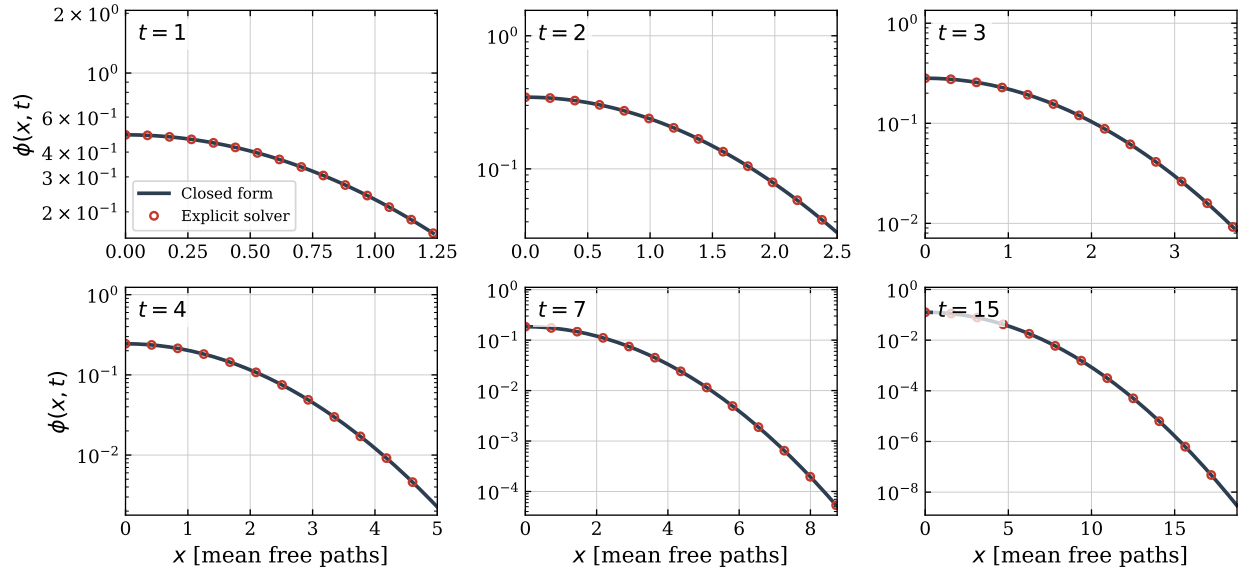


Figure 6: Part 3(c): the explicit solver (markers, thinned) over the closed-form diffusion Green's function it discretises (solid), at $c = 1$ and $N = 2000$. The two are indistinguishable at this scale over eight decades of flux. What separates them is Figure 7.

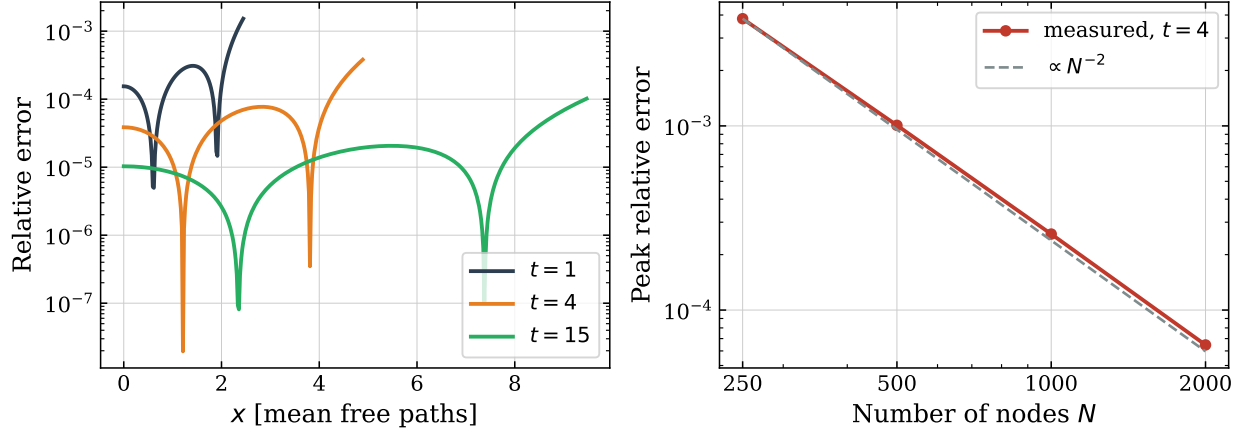


Figure 7: Part 3(c): the solver's relative error against the closed form, within three standard deviations. Left, across the domain at three times. Right, the peak error at $t = 4$ against the node count, against a second-order guide.

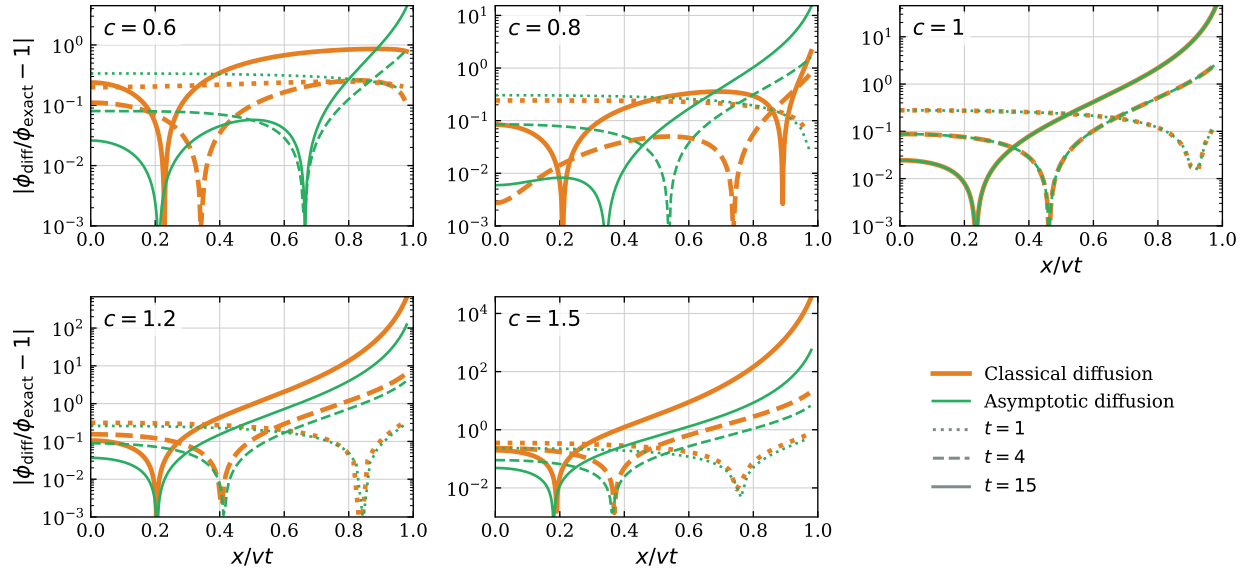


Figure 8: Part 3(d): relative error (43) of the two numerical diffusion solutions against the exact transport flux, one panel per c , against the scaled position x/vt . The causal front is at 1. At $c = 1$ the two diffusion coefficients coincide.